Technical Appendix: Top-k Sparsification for Enhancing Privacy-Utility Tradeoff in AirComp Federated Learning

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Appendix A
Algorithm Flow and Assumptions

To facilitate the analysis, we first formally define the algorithm flow, error feedback mechanism, and core assumptions. Let $r = k/d$ be the sparsification ratio. All vectors are denoted in boldface (e.g., $\boldsymbol{\theta}$, $\mathbf{e}$, $\boldsymbol{\Delta}$).

A. Algorithm Flow and Error Feedback Mechanism

Following the main text, the raw accumulated local update after τ SGD steps is

$$\boldsymbol{\Delta}_i^t \triangleq \mathbf{u}_i^t = -\eta \sum_{s=1}^{\tau} \mathbf{g}_i^{t,s-1}, \tag{1}$$

where $\mathbf{g}_i^{t,s-1}$ is the stochastic gradient satisfying Assumption 2. To compensate for the bias introduced by top-k compression and systemic clipping, each device maintains a local error memory $\mathbf{e}_i^t$ initialized as $\mathbf{e}_i^0 = \mathbf{0}$. The "sparsification-then-clipping" feedback mechanism proceeds as follows:

1) **Compensation:** Form the ideal update $\mathbf{v}_i^t = \boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1}$.
2) **Sparsification:** Extract the sparse signal $\tilde{\mathbf{v}}_i^t = \mathrm{Top}_k(\mathbf{v}_i^t)$.
3) **Error Update:** The unselected residual is stored: $\mathbf{e}_i^t = \mathbf{v}_i^t - \tilde{\mathbf{v}}_i^t$.
4) **Systemic Clipping:** The sparse signal is clipped for transmission: $\mathbf{s}_i^t = \mathrm{clip}(\tilde{\mathbf{v}}_i^t, \eta\tau C)$.

Defining the systemic clipping bias as $\boldsymbol{\xi}_i^t \triangleq \tilde{\mathbf{v}}_i^t - \mathbf{s}_i^t$, we obtain the exact algebraic relationship

$$\mathbf{s}_i^t = \tilde{\mathbf{v}}_i^t - \boldsymbol{\xi}_i^t = \mathbf{v}_i^t - \mathbf{e}_i^t - \boldsymbol{\xi}_i^t = \boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1} - \mathbf{e}_i^t - \boldsymbol{\xi}_i^t. \tag{2}$$

B. Core Assumptions

Assumption 1 (Smoothness and Lower Bound). *Each local objective function $f_i : \mathbb{R}^d \to \mathbb{R}$ is L-smooth, i.e., $\|\nabla f_i(\boldsymbol{\theta}) - \nabla f_i(\boldsymbol{\theta}')\|_2 \le L\|\boldsymbol{\theta} - \boldsymbol{\theta}'\|_2$ for all $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathbb{R}^d$. The global objective satisfies $f(\boldsymbol{\theta}) \ge f^* > -\infty$.*

Assumption 2 (Unbiased Gradient and Bounded Variance). *The local stochastic gradient is an unbiased estimator with bounded variance: $\mathbb{E}_t[\mathbf{g}_i^{t,s}] = \nabla f_i(\boldsymbol{\theta}_i^{t,s-1})$, $\mathbb{E}_t[\|\mathbf{g}_i^{t,s} - \nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\|_2^2] \le \zeta_i^2$. We denote $\zeta^2 = \frac{1}{N}\sum_{i=1}^N \zeta_i^2$.*

Assumption 3 (Bounded Heterogeneity). *There exist constants $G^2 \ge 1$ and $\kappa^2 \ge 0$ such that $\frac{1}{N}\sum_{i=1}^N \|\nabla f_i(\boldsymbol{\theta})\|_2^2 \le G^2\|\nabla f(\boldsymbol{\theta})\|_2^2 + \kappa^2$.*

Assumption 4 (Bounded Clipping Bias). *The systemic clipping bias $\boldsymbol{\xi}_i^t \triangleq \tilde{\mathbf{v}}_i^t - \mathbf{s}_i^t$ satisfies*

$$\mathbb{E}_t\|\boldsymbol{\xi}_i^t\|_2^2 \le \eta^2\tau^2\beta^2, \tag{3}$$

where β^2 bounds the truncation residual.

Appendix B
Auxiliary Lemmas

Lemma 1 (Contraction Property of Top-k). *For any input vector $\mathbf{v} \in \mathbb{R}^d$, the deterministic compression error satisfies:*

$$\|\mathbf{v} - \mathrm{Top}_k(\mathbf{v})\|_2^2 \le (1-r)\|\mathbf{v}\|_2^2. \tag{4}$$

Proof. Let the coordinates of $\mathbf{v}$ be sorted by magnitude such that $v_{(1)}^2 \ge v_{(2)}^2 \ge \cdots \ge v_{(d)}^2$. The error vector $\mathbf{v} - \mathrm{Top}_k(\mathbf{v})$ consists exactly of the smallest $d-k$ elements. Because these elements have the smallest magnitudes, their average energy is inherently bounded by the average energy of the entire vector:

$$\frac{1}{d-k}\sum_{j=k+1}^{d} v_{(j)}^2 \le \frac{1}{d}\sum_{j=1}^{d} v_{(j)}^2 = \frac{1}{d}\|\mathbf{v}\|_2^2.$$

Multiplying both sides by $d-k$, we obtain $\|\mathbf{v}-\mathrm{Top}_k(\mathbf{v})\|_2^2 \le \frac{d-k}{d}\|\mathbf{v}\|_2^2 = (1-r)\|\mathbf{v}\|_2^2$. □

Lemma 2 (**Error Contraction Step**). *Given* $\mathbf{v}_i^t = \mathbf{\Delta}_i^t + \mathbf{e}_i^{t-1}$ *and* $\mathbf{e}_i^t = \mathbf{v}_i^t - \mathrm{Top}_k(\mathbf{v}_i^t)$,

$$\mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 \le (1-r)\mathbb{E}_t\|\mathbf{\Delta}_i^t + \mathbf{e}_i^{t-1}\|_2^2. \tag{5}$$

Proof. Trivial from taking the expectation on both sides of the deterministic property in Lemma 1 and applying the definition of $\mathbf{v}_i^t$. □

Lemma 3 (**Bounded Local Divergence**). *Under Assumptions 1–3, the physical local model drift satisfies*

$$\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\boldsymbol{\theta}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 \le 16\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 16\eta^2\tau^2\kappa^2 + 4\tau\eta^2\zeta^2. \tag{6}$$

Proof. According to the local update rule $\boldsymbol{\theta}_i^{t,h} = \boldsymbol{\theta}_i^{t,h-1} - \eta\mathbf{g}_i^{t,h-1}$ (where $h \ge 1$ denotes the local step iteration) and the initial synchronization $\boldsymbol{\theta}_i^{t,0} = \boldsymbol{\theta}^t$, applying the Peter-Paul inequality $\|\mathbf{a}+\mathbf{b}\|_2^2 \le (1+\frac{1}{2\tau-1})\|\mathbf{a}\|_2^2 + 2\tau\|\mathbf{b}\|_2^2$ yields:

$$\begin{aligned}
\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h} - \boldsymbol{\theta}^t\|_2^2 &= \mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t - \eta\mathbf{g}_i^{t,h-1}\|_2^2 \\
&\le \Big(1+\frac{1}{2\tau-1}\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 2\tau\eta^2\mathbb{E}_t\|\mathbf{g}_i^{t,h-1}\|_2^2 \\
&\le \Big(1+\frac{1}{2\tau-1}\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\tau\eta^2\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}_i^{t,h-1}) - \nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + 4\tau\eta^2\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + \eta^2\zeta_i^2 \\
&\le \Big(1+\frac{1}{2\tau-1} + 4\eta^2L^2\tau\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\eta^2\tau\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + \eta^2\zeta_i^2.
\end{aligned}$$

Averaging over all clients $i \in [N]$, applying the dissimilarity Assumption 3 for G^2 and κ^2, and enforcing the learning rate condition $\eta \le \frac{1}{2\sqrt{2}\tau L}$ ensures $4\eta^2L^2\tau \le \frac{1}{2\tau} \le \frac{1}{2\tau-1}$. With $\zeta^2 = \frac{1}{N}\sum_i \zeta_i^2$, we obtain:

$$\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h} - \boldsymbol{\theta}^t\|_2^2 \le \Big(1+\frac{1}{\tau-1}\Big)\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2.$$

Unrolling the recursion for the local model after $s-1$ updates (evaluating from $h=0$ to $s-2$, with $\boldsymbol{\theta}_i^{t,0} = \boldsymbol{\theta}^t$), we get:

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\boldsymbol{\theta}_i^{t,s-1} - \boldsymbol{\theta}^t\|_2^2 &\le \sum_{h=0}^{s-2}\Big(1+\frac{1}{\tau-1}\Big)^h\Big[4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2\Big] \\
&\le (\tau-1)\left[\Big(1+\frac{1}{\tau-1}\Big)^\tau - 1\right] \times \Big[4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2\Big] \\
&\le 16\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 16\eta^2\tau^2\kappa^2 + 4\tau\eta^2\zeta^2,
\end{aligned} \tag{7}$$

where the last inequality follows from bounding the geometric series and applying $(1+\frac{1}{\tau-1})^\tau \le 5$ (for $\tau > 1$). This holds for any $s \le \tau$, completing the proof. □

Lemma 4 (**Error Memory Bound**). *The global average of the error memory is bounded by*

$$\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 \le \mu\frac{1}{N}\sum_{i=1}^N \mathbb{E}_t\|\mathbf{\Delta}_i^t\|_2^2, \tag{8}$$

where $\mu = \frac{2(1-r)(2-r)}{r^2}$.

Proof. Using the Peter-Paul inequality $\|\mathbf{a}+\mathbf{b}\|_2^2 \le (1+\alpha)\|\mathbf{a}\|_2^2 + (1+\alpha^{-1})\|\mathbf{b}\|_2^2$ with Lemma 2, we choose $\alpha = \frac{r}{2(1-r)}$ to obtain contraction factor $\rho = (1-r)(1+\alpha) = \frac{2-r}{2} < 1$. Unrolling the recursion yields the constant $\mu = \frac{\beta}{1-\rho} = \frac{2(1-r)(2-r)}{r^2}$. □

Appendix C
Proof of Main Convergence Bound (Theorem 2)

A. Construction of the Virtual Sequence and Separation

Define the global error memory $\bar{\mathbf{e}}^{t-1} \triangleq \frac{1}{N}\sum_{i=1}^N \mathbf{e}_i^{t-1}$ and the virtual global model $\tilde{\boldsymbol{\theta}}^t = \boldsymbol{\theta}^t + \bar{\mathbf{e}}^{t-1}$, which completely absorbs the unsent Top-k errors. From the physical update rule $\boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t + \frac{1}{N}\sum_{i=1}^N \mathbf{s}_i^t + \frac{\mathbf{z}^t}{Nb_t}$ and the exact structural relation (2), the error feedback naturally cancels out, yielding the virtual recursion:

$$\tilde{\boldsymbol{\theta}}^{t+1} = \tilde{\boldsymbol{\theta}}^t + \frac{1}{N}\sum_{i=1}^N \mathbf{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t + \frac{\mathbf{z}^t}{Nb_t}, \tag{9}$$

where $\bar{\boldsymbol{\xi}}^t = \frac{1}{N}\sum_{i=1}^N \boldsymbol{\xi}_i^t$. Applying the L-smoothness descent lemma yields the separation

$$\mathbb{E}_t[f(\tilde{\boldsymbol{\theta}}^{t+1}) - f(\tilde{\boldsymbol{\theta}}^t)] \leq \underbrace{\mathbb{E}_t\langle \nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t\rangle}_{T_1} + \underbrace{\frac{L}{2}\mathbb{E}_t\Big\|\frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t + \frac{\mathbf{z}^t}{Nb_t}\Big\|_2^2}_{T_2}. \tag{10}$$

B. Bounding T_1 and T_2

1) Descent Direction T_1: Substituting $\boldsymbol{\Delta}_i^t = -\eta\sum_{s=1}^\tau \mathbf{g}_i^{t,s-1}$ and applying the unbiasedness of gradients (Assumption 2), the main inner product expands as:

$$\mathbb{E}_t\langle \nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t\rangle = -\eta\tau\mathbb{E}_t\langle \nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\rangle.$$

Using the inner product identity $2\langle a,b\rangle = \|a\|^2 + \|b\|^2 - \|a-b\|^2$, this becomes:

$$-\frac{\eta\tau}{2}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 - \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\Big\|_2^2 + \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\big(\nabla f(\tilde{\boldsymbol{\theta}}^t) - \nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\big)\Big\|_2^2.$$

Applying Jensen's inequality and L-smoothness to the last term yields the heterogeneity drift $\frac{\eta L^2}{2N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2$. For the separate clipping bias term, applying the Peter-Paul inequality $-\langle a,b\rangle \leq \frac{\eta\tau}{4}\|a\|^2 + \frac{1}{\eta\tau}\|b\|^2$ alongside Assumption 4 guarantees:

$$-\mathbb{E}_t\langle \nabla f(\tilde{\boldsymbol{\theta}}^t), \bar{\boldsymbol{\xi}}^t\rangle \leq \frac{\eta\tau}{4}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{1}{\eta\tau}(\eta^2\tau^2\beta^2) = \frac{\eta\tau}{4}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \eta\tau\beta^2.$$

Summing the expanded terms, the primary gradient coefficient becomes $-\frac{\eta\tau}{2} + \frac{\eta\tau}{4} = -\frac{\eta\tau}{4}$, which we safely relax to $-\frac{\eta\tau}{8}$ to ensure ample descent margin. Hence:

$$T_1 \leq -\frac{\eta\tau}{8}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{\eta L^2}{2N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 - \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\Big\|_2^2 + \eta\tau\beta^2. \tag{11}$$

2) Variance and Noise T_2: Separating the zero-mean channel noise ($\mathbb{E}_{z^t}[\mathbf{z}^t] = \mathbf{0}$) and sequentially applying the basic inequality $\|\mathbf{a}-\mathbf{b}\|_2^2 \leq 2\|\mathbf{a}\|_2^2 + 2\|\mathbf{b}\|_2^2$, Jensen's inequality, and Assumption 4, we obtain:

$$T_2 \leq L\frac{1}{N}\sum_i \mathbb{E}_t\|\boldsymbol{\Delta}_i^t\|_2^2 + L\eta^2\tau^2\beta^2 + \frac{L\sigma_0^2 d}{4N^2 b_t^2}. \tag{12}$$

Expanding $\mathbb{E}_t\|\boldsymbol{\Delta}_i^t\|_2^2$ completely via Assumptions 2 and 3 yields the explicit form:

$$T_2 \leq 2L\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{2L^3\eta^2\tau}{N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 + L\eta^2\tau^2(2\kappa^2+\zeta^2) + L\eta^2\tau^2\beta^2 + \frac{L\sigma_0^2 d}{4N^2 b_t^2}. \tag{13}$$

C. Bounding the Virtual Local Divergence

Define the local divergence toward the virtual model as $\Psi_t \triangleq \frac{1}{N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2$. Using the algebraic decomposition $\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1} = \bar{\mathbf{e}}^{t-1} + (\boldsymbol{\theta}^t - \boldsymbol{\theta}_i^{t,s-1})$ and Lemma 3, together with the gradient conversion bound $\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \leq 2\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2L^2\|\bar{\mathbf{e}}^{t-1}\|_2^2$, we obtain:

$$\Psi_t \leq 64\eta^2\tau^3G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2\tau(1+32\eta^2\tau^2G^2L^2)\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2 + 32\eta^2\tau^3\kappa^2 + 8\tau^2\eta^2\zeta^2. \tag{14}$$

D. Learning Rate Conditions and Single-Step Descent

Combining T_1 and T_2, substituting Ψ_t, and safely bounding the global error memory $\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$ via Lemma 4 (using Jensen's inequality, $\|\bar{\mathbf{e}}^{t-1}\|_2^2 \leq \frac{1}{N}\sum_{i=1}^N\|\mathbf{e}_i^{t-1}\|_2^2$), the total positive gradient penalty accumulates to:

$$C_\nabla = 2L\eta^2\tau^2G^2 + \eta L^2\Big(\frac{1}{2} + 2L\eta\tau\Big)\Big[64\eta^2\tau^3G^2 + 4\mu\eta^2\tau^3G^2(1+32\eta^2\tau^2G^2L^2)\Big]. \tag{15}$$

For guaranteed convergence, we strictly enforce $C_\nabla \leq \frac{\eta\tau}{16}$, which decomposes into the following sufficient conditions:

1) $2L\eta\tau G^2 \leq \frac{1}{32} \;\Rightarrow\; \eta \leq \frac{1}{64L\tau G^2}$,
2) $\frac{1}{2} + 2L\eta\tau \leq 1 \;\Rightarrow\; \eta \leq \frac{1}{4L\tau}$,
3) $32\eta^2\tau^2G^2L^2 \leq 0.5 \;\Rightarrow\; \eta \leq \frac{1}{8\tau GL}$,
4) $L^2\eta^2\tau^2G^2(64+6\mu) \leq \frac{1}{32} \;\Rightarrow\; \eta \leq \frac{1}{\tau GL\sqrt{2048+192\mu}}$.

Noting that $G \geq 1$ holds by Assumption 3, it is straightforward to verify that $\frac{1}{64L\tau G^2} < \min\left\{\frac{1}{4L\tau}, \frac{1}{8\tau GL}\right\}$. Therefore, conditions 2 and 3 are always strictly subsumed by condition 1. Consequently, the effective learning rate is governed purely by the bottleneck of conditions 1 and 4, which directly yields the simplified two-term constraint presented in the main text:

$$\eta \leq \min\left\{\frac{1}{64L\tau G^2}, \frac{1}{\tau GL\sqrt{2048+192\mu}}\right\}.$$

Under these exact constraints, the one-round descent formula simplifies elegantly to:

$$\mathbb{E}_t[f(\tilde{\boldsymbol{\theta}}^{t+1}) - f(\tilde{\boldsymbol{\theta}}^t)] \leq -\frac{\eta\tau}{8}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \eta^2\tau^2 LC_{\text{var}}(\kappa^2+\zeta^2) + \mathcal{E}_{\text{clip}} + \frac{L\sigma_0^2 d}{4N^2 b_t^2}, \tag{16}$$

where $\mathcal{E}_{\text{clip}} \triangleq (\eta\tau + L\eta^2\tau^2)\beta^2$, and the intrinsic heterogeneity constant C_{var} is defined precisely as (assuming $\kappa^2+\zeta^2 > 0$; otherwise variance identically vanishes):

$$C_{\text{var}} \triangleq \frac{2\kappa^2+\zeta^2}{\kappa^2+\zeta^2} + \eta L\Big[\frac{32\tau\kappa^2+8\zeta^2}{\kappa^2+\zeta^2} + 2\tau\mu(1+32\eta^2\tau^2G^2L^2)\frac{2\kappa^2+\zeta^2}{\kappa^2+\zeta^2}\Big]. \tag{17}$$

E. Virtual Model Convergence

Telescoping the sum over $t = 0, \ldots, T-1$ and recognizing $f(\boldsymbol{\theta}^0) = f(\tilde{\boldsymbol{\theta}}^0)$, we define the initial distance $\Delta f \triangleq f(\boldsymbol{\theta}^0) - f^*$. Rearranging the equation yields the tight virtual bound:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 \leq \frac{8\Delta f}{T\eta\tau} + 8\eta\tau LC_{\text{var}}(\kappa^2+\zeta^2) + \frac{8\mathcal{E}_{\text{clip}}}{\eta\tau} + \frac{2L\sigma_0^2 d}{T\eta\tau N^2}\sum_{t=0}^{T-1}\frac{1}{b_t^2}. \tag{18}$$

F. Conversion to the Physical Model

From the conversion $\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \leq 2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2L^2\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$, we explicitly define the error sequence $E_{t-1} \triangleq 2L^2\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$. Invoking Lemma 4 (with Jensen's inequality) and substituting the exact bound of Ψ_{t-1}, we establish a rigorous recursive contraction relation for the error sequence:

$$E_{t-1} \leq 8L^2\mu\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^{t-1})\|_2^2 + 3L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2) + \rho E_{t-2}, \tag{19}$$

where the contraction factor is defined as $\rho \triangleq 6L^2\mu\eta^2\tau^2$. The rigorous learning rate constraints derived above strictly guarantee $\rho \leq \frac{1}{32} \leq 0.5$. Unrolling this recursion over T rounds and applying the bounding logic for geometric series sum $\frac{1}{1-\rho} \leq 2$, we eliminate the recursive term:

$$\begin{aligned}\sum_{t=0}^{T-1}E_{t-1} &\leq \frac{1}{1-\rho}\sum_{t=0}^{T-1}\Big(8L^2\mu\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 3L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2)\Big)\\ &\leq 16L^2\mu\eta^2\tau^2G^2\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 6L^2\mu\eta^2\tau^2T(2\kappa^2+\zeta^2).\end{aligned} \tag{20}$$

Substituting this sequence sum back gives the physical gradient amplification factor $\Gamma \triangleq 2 + 16L^2\mu\eta^2\tau^2G^2$. Scaling (18) entirely by Γ and correctly assigning the residual error-delay constants produces the final physical convergence bound:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \leq \frac{2\Gamma L\sigma_0^2 d}{T\eta\tau N^2}\sum_{t=0}^{T-1}\frac{1}{b_t^2} + \Phi(k), \tag{21}$$

$$\Phi(k) \triangleq \frac{8\Gamma\Delta f}{T\eta\tau} + \frac{8\Gamma\mathcal{E}_{\text{clip}}}{\eta\tau} + 8\Gamma\eta\tau LC_{\text{var}}(\kappa^2+\zeta^2) + 6L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2), \tag{22}$$

which completes the proof of Theorem 2.